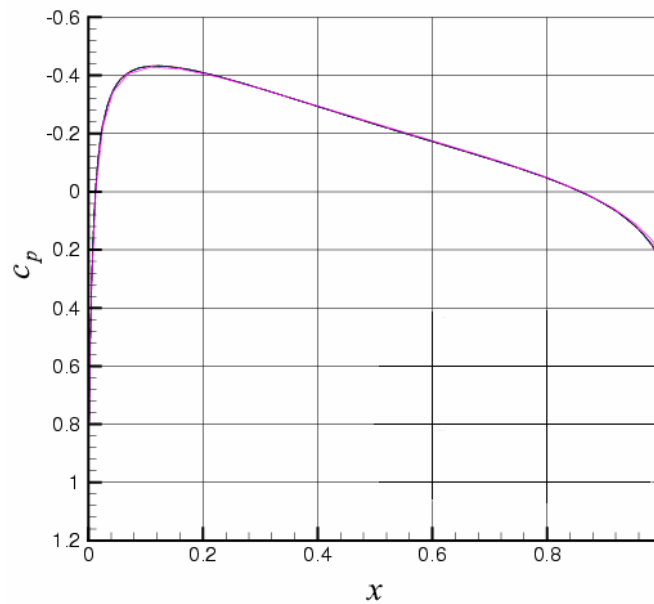


Homework 10

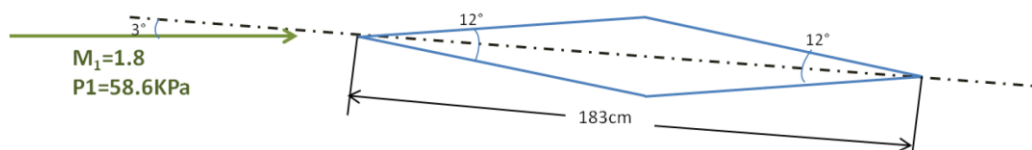
1-The distribution of the pressure coefficient at $M=0.3$ at zero angle of attack over NACA 0012 is shown in figure below. Find the critical Mach number. Use Laiton relation. Write a MATLAB code to solve this problem, i.e, use fsolve function, you can use AI.



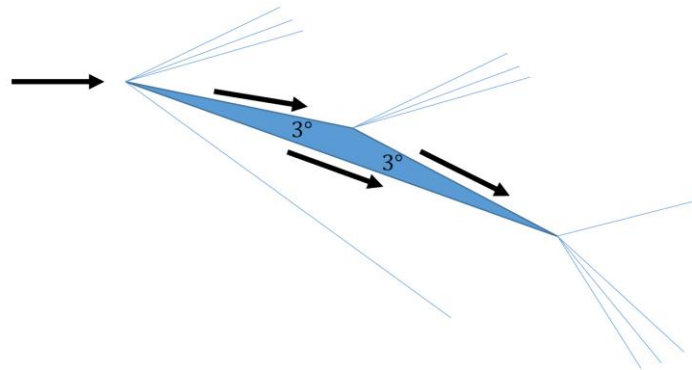
$$C_{p_{cr}} = \frac{2}{\gamma M_{cr}^2} \left[\left(\frac{1 + \frac{\gamma-1}{2} M_{cr}^2}{1 + \frac{\gamma-1}{2}} \right)^{\frac{\gamma}{\gamma-1}} - 1 \right]$$

$$C_p = \frac{C_{p_0}}{\sqrt{1 - M_\infty^2} + \frac{M_\infty^2}{\sqrt{1 - M_\infty^2}} \frac{C_{p_0}}{2} \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right)}$$

2- The following symmetrical airfoil is at $M=1.8$, $P=58.6$ KPa and angle of attack of 3 degree. Use the shock/expansion method and linearized theory to find C_l and C_d .



3- Consider a half-diamond airfoil with a freestream Mach number of $M = 3.7$, a wedge angle of $\epsilon = 3.0$, and an angle of attack of $\alpha = 4.0$. Compute the pressure ratio for each surface, the lift coefficient, and the drag coefficient use the shock/expansion method and linearized theory.



4- Consider the trapezoidal airfoil shown in the diagram above. The maximum thickness is t , the length from tip to tip is c , and angles at the leading and trailing edges are both ϵ . Regions of uniform flow over the airfoil are numbered 1 through 5. The angle of attack α is measured relative to the flat, bottom side of the airfoil.

(a) Classify the waves at the boundaries of each region (Wave 1-2, Wave 2-3, Wave 3-4, and Wave 1-5) as an expansion or a shock according to the angle of attack α . Make a table like this, replacing the “?” marks with the appropriate angle:

Wave	Shock Occurs	Expansion Occurs
1-2	$? < \alpha < ?$	$? < \alpha < ?$
2-3	$? < \alpha < ?$	$? < \alpha < ?$
3-4	$? < \alpha < ?$	$? < \alpha < ?$
1-5	$? < \alpha < ?$	$? < \alpha < ?$

“Never” and “always” are acceptable entries in the table.

(b) Using shock-expansion theory, compute the lift and drag coefficients for $M = 3.0$, $\alpha = 3.5^\circ$, $\epsilon = 2^\circ$, and $t/c = 0.03$.

